Assignment 4

DAY 1: introduction to network hardware Theory (1 hour): Overview of modems, router, switches, access points, and network cables.

Task 1: Labeled Diagram of a Typical Home Wi-Fi Setup

A typical home Wi-Fi network uses different networking devices to connect computers, smartphones and other devices to the Internet.

A basic home network can be arranged as:

Internet → Modem → Router → Switch → Wired Devices

The **router** can also provide Wi-Fi directly. An **access point** can be connected to the network to extend wireless coverage to areas where the router's Wi-Fi signal is weak.

Network Flow

Internet/ISP
↓
Modem
↓
Router
↓
Switch
↓
Wired Devices

The **Access Point** is connected to the network and provides Wi-Fi coverage. A **laptop and smartphone** can connect wirelessly to the router or access point.



Task 2: Functions of Networking Hardware

Device	Main Function	Real-Life Example
Modem	Connects the local network to the Internet service provided by the ISP and converts/signals data as required by the access technology.	Used at home to connect an ISP connection such as cable, DSL or fiber equipment.
Router	Routes IP packets between different networks and usually provides local network and Wi-Fi functions.	Used at home to share one Internet connection with phones, laptops and TVs.
Switch	Connects multiple wired devices within the same local network and forwards Ethernet frames to the appropriate device.	Used in an office to connect desktop computers, printers and servers.
Access Point	Provides wireless network access and can extend Wi-Fi coverage.	Used in offices, schools and large homes to provide Wi-Fi in additional areas
Network Cable	Carries network data between wired devices.	Ethernet cable used to connect a PC to a router or switch.



Task 3: Types of Network Cables

1. Ethernet Cable (RJ45)

Description:
Ethernet cables are commonly used to connect computers, routers, switches, printers and other network devices in a **LAN (Local Area Network)**.

Common use:
Used for wired Internet and LAN connections in homes, offices and computer labs.



2. Coaxial Cable

Description:
Coaxial cables have a central conductor with shielding around it. They are commonly used for **cable television, cable Internet and some broadband connections**.

Common use:
Used to connect a cable modem or television service to the provider's cable network.



3. Fiber-Optic Cable

Description:
Fiber-optic cables transmit data using **light through optical fibers**. They provide high bandwidth and are suitable for long-distance communication.

Common use:
Used for high-speed Internet, telecommunications, data centers and backbone networks.



Task 4: Router vs Switch Comparison

Feature	**Router**	**Switch**
Main Use	Connects different networks and routes IP packets between them.	Connects multiple devices within the same local network.
Typical Location	Usually placed near the Internet connection in homes and offices.	Used in homes, offices, labs and server rooms to connect wired devices.
How It Handles Data	Uses IP addresses to decide where packets should go.	Uses MAC addresses to forward Ethernet frames within a LAN.
Ports	Usually has a WAN/Internet port plus several LAN ports.	Usually has multiple Ethernet ports for LAN devices.
Wi-Fi	Many home routers provide Wi-Fi.	A normal Ethernet switch does not provide Wi-Fi.
Internet Connection	Can connect the local network to the Internet.	Normally does not directly connect different networks to the Internet
Example	Home Wi-Fi router connecting phones and laptops to the Internet.	Office switch connecting multiple desktop computers.



Figure 6: Router



Figure 7: Switch